

# CANaryNode Power Tree

**What this document is:** A map of every voltage rail on the board, what creates it, what consumes it, and how much current each load draws. Used to verify that the chosen regulator is correctly sized, decoupling is adequate, and trace widths are sufficient.

**Design summary:** The board has two rails. USB-C VBUS supplies 5V from the host (up to 3A per USB 2.0 spec). That 5V feeds the TPS62160 buck converter, which steps it down to 3.3V at up to 1A. Everything else on the board — MCU, both sensors, CAN transceiver, LEDs — runs from that single 3.3V rail. The 5V rail is a pass-through with no other loads.

A buck converter was chosen over an LDO for efficiency (~90% vs ~66%). At expected loads this difference is small in absolute terms but demonstrates correct power-supply selection for portfolio purposes and scales cleanly if future revisions add higher-current peripherals.

## What's on the power tree (current-drawing chips):

1. STM32G474
2. ICM-42688-P
3. BME280
4. TCAN334
5. (LEDs — not a chip but draws current)

## What's NOT on the power tree, but IS going on the PCB:

- **USBLC6-2SC6 ESD chip** — passive TVS diode, draws essentially zero current in normal operation (only conducts during ESD events). Not a load.
- **TPS62160 buck converter** — this is the *source* of the 3.3V rail, not a load on it. Its own quiescent current is ~17  $\mu$ A, negligible.
- **All passives** — resistors, capacitors, inductor. They don't have a "supply current" the way active chips do. Their current is determined by the circuit they're in (e.g., LED resistor current is already counted in the LED row).
- **Pull-up resistors on I2C** (4.7 k $\Omega$ ): each pulls ~700  $\mu$ A when the bus is low. Tiny but not zero — we'll handle as part of MCU GPIO budget.
- **Connectors** — USB-C, CAN screw terminal, SWD/UART headers. Mechanical only, no current.
- **LEDs** — they draw current but aren't "chips" per se. Counted as a row already.
- **Button** — passive switch, no current draw except briefly when pressed.

**\*\* For 'load name' we are using the operating mode of the chip which would draw the worst case current.**

**STM32G474 active** — CPU in Run mode at 170 MHz with USB, FDCAN, SPI, I2C, USART all enabled. Worst-case ongoing current.

**ICM-42688-P normal** — Accelerometer + gyroscope both on in Low Noise mode. Continuous 6-axis sampling.

**BME280 forced mode** — Take one measurement on command, return to sleep. The mode used for slow-changing environmental data; current dominated by brief measurement spikes.

**TCAN334 active** — Normal mode, actively driving the bus (dominant/TX state). Higher than receive-only current.

**LEDs (4× @ ~5mA all on)** — Worst case where all 4 status LEDs are lit simultaneously. 5mA per LED is a design choice (bright enough to see, low enough to ignore in power budget).

- "Typical" = current in the mode you expect during normal operation
- "Max" = current in the highest-power mode the chip supports

| Load                                                                         | Typical (mA)     | Max (mA) | Source/Datasheet Section                   |
|------------------------------------------------------------------------------|------------------|----------|--------------------------------------------|
| STM32G474 active @170 MHz                                                    | 29.5             | 42.0     | DS12288 §5.3.5, Table 21, 170 MHz row      |
| ICM-42688-P normal Low-Noise Mode, 6-Axis Gyroscope + Accelerometer          | 0.88             | 1.5(est) | DS-000347 §3.3.1, Table 3, 6-axis LN       |
| BME280 forced mode: 714 $\mu$ A (I_DDP, current during pressure measurement) | 0.71             | 1.0(est) | BST-BME280-DS002 §1.1, Table 1, I_DDP peak |
| TCAN334 active Dominant, High Bus Load                                       | 15.0(est)        | 60.0     | SLLSEU4 §5.5, I_CC Normal Mode             |
| LEDs (4× @ ~5mA all on)                                                      | 5.0 (1.0 on avg) | 20.0     | Design choice                              |
| **Subtotal**                                                                 | 51.1             | 124.5    | —                                          |
| **+ 50% headroom**                                                           | —                | 186.8    | —                                          |
| **TPS62160 rating**                                                          | —                | 1000     | —                                          |

**\*\*TPS62160(Buck Converter) Rating" explained**

- EVERYTHING on the board is connected to the 3.3V rail (for CANary node), the buck needs to be able to provide all the current everything is pulling. The buck provides 1000mA we are only using ~187mA at most.

**I picked a 1A buck for a 200 mA load on purpose. It runs cooler, leaves room to add features later, handles sudden current spikes better, and the cheaper alternative only saves 50 cents — not worth the tradeoff.**

The rating is the regulator's maximum output current — how much it can deliver before it fails or shuts down. For the TPS62160 it's 1000 mA (1 A) per its datasheet.

You compare your headroomed worst case against this number:

**186.8 mA needed vs 1000 mA available = 18.7% utilization**

You're using less than a fifth of what the regulator can deliver. Massive headroom.

## **Note for power-tree doc — STM32G474 current sourcing**

Table used: Table 21 — "Current consumption in Run and Low-power run modes, code with data processing running from Flash in single Bank, ART enable (Cache ON Prefetch OFF)" (DS12288, §5.3.5)

Why this table out of the ~14 current-consumption tables:

- Run mode (not Low-power run): worst-case sizing — assume CPU is always at full clock
- Code from Flash, single bank: the default execution configuration. Dual-bank is only enabled for A/B firmware update schemes (relevant to Project 2 bootloader, not CANaryNode)
- ART enable, Cache ON Prefetch OFF: ST's standard "typical use" condition — most realistic for everyday firmware
- Sleep/Stop/Standby tables ignored — those are for low-power analysis, not regulator sizing

Row picked: f\_HCLK = 170 MHz (max clock for G474), all peripherals enabled. This is the worst-case ongoing current the firmware will demand from the 3.3V rail.

Currents read at Typ = 25°C, Max = 85°C. Selected 85°C as the upper bound for typical indoor/lab operation. If CANaryNode were deployed in a vehicle or outdoor environment, the 125°C column would be more appropriate.

---

## **Note for power-tree doc — ICM-42688-P current sourcing**

Table used: Table 3 — D.C. Electrical Characteristics (DS-000347, §3.3.1)

**Row picked:** Low-Noise Mode, 6-Axis Gyroscope + Accelerometer = 0.88 mA typical at 25°C. This is the worst-case ongoing current — both sensors active in highest-performance mode.

**Caveats captured:**

Datasheet characterized at VDD = 1.8 V. CANaryNode runs the chip at 3.3 V, so actual current may be 10–20% higher. Will measure during bring-up.

Datasheet provides **Typ only, no Max column. Used 1.5 mA** as estimated Max for sizing headroom.

**Filed away for schematic time (not power-related):**

I<sup>2</sup>C address from §3.3.2 Table 4: 0x68 (AP\_AD0 = 0) or 0x69 (AP\_AD0 = 1). Pin-strap choice during schematic capture.

VDDIO and VDD can be separate rails (chip supports split supplies). We'll tie both to 3.3V for simplicity.

Spec'd operating range: -40°C to +85°C — fine for indoor/lab use.

---

## **Note for power-tree doc — BME280 current sourcing**

**Table used:** Table 1 — Electrical parameter specification (BST-BME280-DS002 §1.1)

**Number picked:** 714 µA (I<sub>DDP</sub>, current during pressure measurement) — the highest single-measurement current in the table.

**Reasoning:**

- BME280 measures temperature, humidity, and pressure **sequentially**, not simultaneously. Peak current = highest single measurement, not the sum.
- Pressure measurement is the worst case at 714 µA. Humidity is 340 µA, temperature is 350 µA.
- In forced mode, the chip is in standby (0.2 µA) most of the time and only spikes to 714 µA briefly per measurement. **Average current is much lower than peak**, but the regulator must handle peak instantaneous demand → power tree uses peak.

**Caveats captured:**

- Datasheet characterized at VDD = 1.8 V. CANaryNode runs the chip at 3.3 V — actual current may be slightly higher.
- Datasheet provides Typ only for measurement currents. Max value (1.0 mA) estimated for headroom.

## **Note for power-tree doc — TCAN334 current sourcing**

**Table used:** Electrical Characteristics, I<sub>CC</sub> row (TCAN334 datasheet SLLSEU4, §5.5)

**Numbers picked:**

**Max:** 60 mA — Dominant state, High Bus Load (50Ω). Represents a properly terminated CAN bus (two 120Ω resistors in parallel = 60Ω, close enough). Worst case for active transmission. Typical: 15 mA (est.) — datasheet shows Recessive state at only 3.5 mA. The bus sits recessive most of the time (idle or transmitting 1-bits), with brief dominant pulses during TX. Estimated average between recessive (3.5 mA) and dominant (60 mA) bursts.

Notes:

TCAN334 datasheet characterized at VCC = 3.3 V — no voltage conversion caveat (unlike the BME280 and IMU).

180 mA "Dominant with bus fault" row ignored — fault condition (CANH shorted to -12V), not normal operation. Buck's overcurrent protection handles this.

## Note for power-tree doc — LEDs

Source: Design choice, not datasheet.

### Numbers picked:

**Max:** 20 mA — 4 LEDs × 5 mA each, all on simultaneously. Worst case where firmware lights every status LED at once.

**Typical:** 5 mA — assumes one LED on at a time on average (e.g., heartbeat blinking, power-on LED steady, others off).

### Sizing reasoning:

5 mA per LED is the design target. Bright enough to see in normal indoor lighting, low enough that 4 of them combined is negligible against the rail budget.

Current set by series resistor:  $R = (V_{\text{rail}} - V_F) / I_{\text{LED}} = (3.3 - \sim 2.0) / 0.005 \approx 260 \Omega$ . Will use standard 330 Ω → ~3.9 mA actual (slightly conservative, fine).

V<sub>F</sub> depends on LED color: red/yellow ≈ 2.0 V, green ≈ 2.2 V, blue/white ≈ 3.0 V. Will pick during schematic capture.